The Stereo Field as a Coordinate System

The Axis

Think of panning as a number line from −100 to +100. Every sound gets an X-coordinate.

Position	−100	−50	0	+50	+100
	Hard L	Left	**Center**	Right	Hard R

Your job: distribute elements so no zone is overcrowded and the energy on both sides is roughly balanced.

The Rules as Axioms

Axiom 1 — Low frequencies anchor at X = 0

Bass and kick live at X = 0. Frequencies below ~80–100 Hz are effectively omnidirectional — wavelengths are too long for your ears to use interaural time difference (ITD), the main spatial localisation mechanism, to determine direction. Panning them off-centre doesn't create a spatial effect; it creates an uneven headroom and phase problem.

> *Bass at X ≠ 0 is a physics constraint, not a stylistic choice.*

Axiom 2 — The centre is load-bearing

X = 0 is a structural column. Overload it and you lose punch and separation. The only elements that must anchor there:

- Kick drum
- Bass / 808
- Lead vocal
- Snare

Everything else should be displaced outward to create space around them.

Axiom 3 — Symmetry ≠ balance

Panning a guitar to −60 and mirroring it at +60 is geometrically symmetric but not automatically balanced. True balance means equal *perceived energy* on both sides — not equal positions. Three quieter elements left and one louder element right can balance perfectly if their RMS levels compensate.

> *Position and loudness are two independent variables.*

Axiom 4 — Width is a function of frequency

The auditory system localises high frequencies more precisely than low ones, via interaural level difference (ILD). This means high-frequency elements can be panned wide without destabilising the mix, while low-mids need to stay closer to centre.

`width = f(pitch)`

Frequency Range	Approx. Hz	Pan Zone
Low	20 – 250 Hz	±0 – 10
Mid	250 Hz – 4 kHz	±20 – 50
High	4 – 20 kHz	±60 – 100

The usable panning range expands as pitch rises — like a triangle.

Axiom 5 — Duplicate elements must diverge

Two identical signals summed at X = 0 add in phase — you get +6 dB of the same thing. Panning them to ±60 instead creates a Haas-effect spread: the signals decorrelate spatially, producing perceived width and fullness without a volume increase. This is the acoustic basis of double-tracking.

> *Two diverged copies > one centred copy.*

The Placement Map

Element	X Position	Notes
Kick drum	0	Never move this
Bass guitar / 808	0	Never move this
Lead vocal	0	Anchor of the song
Snare	0 (or ±5)	Needs to punch centre
Lead synth / piano	0 to ±15	Slight width ok
Claps	±20 to ±40	Spread for width
Hi-hats	±30 to ±60	Alternate L/R if multiple
Rhythm guitar	±50 to ±70	Classic rock: hard pan ±100
Backing vocals	±30 to ±60	Mirror pairs symmetrically
Harmony vocals	±50 to ±80	Wider than backing
Piano (non-lead)	±20 to ±50	
Pads / atmosphere	±60 to ±100	Fill the edges
Sound FX / one-shots	Anywhere	Use movement / automation
Percussion layers	±40 to ±80	Scatter asymmetrically
Strings / orchestra	±30 to ±90	Spread like a real stage

The Frequency-Space Matrix

Think of your mix as a 2D grid — pan position on the horizontal axis, frequency on the vertical.

	Hard L	Left	Center	Right	Hard R
High	Pads	Harmonies	FX	Harmonies	pads
Mid	Guitar	BVox	Lead vocal	BVox	Piano
Low	perc	Clap	Kick/Bass/Snare	Clap	Perc

Goal: no cell empty, no cell overcrowded. Even distribution across both axes — horizontal (L/R) and vertical (frequency). That's a balanced mix.

Here's what it looks like at the end of a session:

Logic Pro mixer — pan positions from an actual session.

Sania Singh: https://saniasingh721.wixsite.com/sania-singh